

Day 07 –IPv4 Addressing & IP Configuration

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THEORY

1. What is a network address?

It identifies the network itself not a specific device and cannot normally be assigned to a device , its the first ip address in the subnet where all the host portion is zeros

2. What is a broadcast address?

It is used to send a packet to all devices within the same subnet , it is the last ip address in a subnet , it cannot be normally be assigned to a device where all the host portion is ones

3. What is a usable host address?

A usable host address is any address from the host portion except the first and the last addresses.

4. What is the first usable IP address?

More than the network address by one.

5. What is the last usable IP address?

Less than broadcast address by one.

6. What does /24 mean?

The prefix length /24 means that the ip is class c and can accept up to 254 hosts (2^8-2)

7. What subnet mask corresponds to /24?

255.255.255.0

8. What is a default gateway?

It is the device that connects a lan to outer networks like the internet most of the time its the router.

9. Why does a router interface need an IP address?

To act like a known default gateway.

10. What is a static IP configuration?

It is the ip configuration that is setup manually and is consistent

11. What does "show ip interface brief" display?

It shows all interfaces used in a cisco router or switch, their ip addresses,administrative status, and their operational protocol status.

12. What does "no shutdown" do conceptually on a Cisco interface?

It enables the interface so the first layer states changes to up in the show ip interface brief.

13. What does interface status "up/up" generally indicate?

The first up shows that the interface is enabled and connected to another device(layer 1 status),and the second up is for layer 2 status (is the protocol used functioning properly)

14. Why should network configuration always be verified?

to ensure that the network operates securely, efficiently, and exactly as intended.

15. Why is IPv4 configuration important in cybersecurity?

Security professionals rely on accurate IP addressing for everything from routing and firewalls to network monitoring, Nmap, Wireshark, and threat hunting. Incorrect addressing can cause connectivity issues, routing errors, monitoring gaps, and security policy mistakes.

IPv4 EXERCISE

For 192.168.100.0/24 identify:

- **Network Address**

192.168.100.0

- **First Usable IP**

192.168.100.1

- **Last Usable IP**

192.168.100.254

- **Broadcast Address**

192.168.100.255

PRACTICAL

From YOUR OWN PC (ipconfig /all) record: IPv4 Address · Subnet Mask · Default Gateway · DNS Server.

Ipv4 address: 192.168.100.209

Subnet mask: 255.255.255.0

Default gateway: 192.168.100.1

DNS server: 8.8.8.8

SCREENSHOTS

1.ipconfig /all

```
PS C:\Windows\System32> ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : LAPTOP-AM22QKJ6
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-16
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::dd09:f6fe:ee7f:b4ac%22(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 789184551
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
NetBIOS over Tcpip. . . . . : Enabled
```

Unknown adapter McAfee VPN:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-74-4B-36-3C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 4:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #4
Physical Address. . . . . : C6-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 5:

```
Media State . . . . . : Media disconnected
```

Wireless LAN adapter Wi-Fi 5:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #5
Physical Address. : CA-13-75-3C-F1-01
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC
Physical Address. : C4-13-75-3C-F1-01
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::e3bb:c8ce:a438:c2e7%5(Preferred)
IPv4 Address. : 192.168.100.209(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained. : Thursday, September 10, 2026 6:48:38 PM
Lease Expires : Saturday, September 12, 2026 3:20:24 PM
Default Gateway : 192.168.100.1
DHCP Server : 192.168.100.1
DHCPv6 IAID : 113513333
DHCPv6 Client DUID. : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
DNS Servers : fe80::1%5
8.8.8.8
8.8.4.4
fe80::f912:e97b:33cc:e623%5
fe80::1%5
NetBIOS over Tcpip. : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : C4-13-75-3C-F1-02
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

2.ping 127.0.0.1

```
PS C:\Windows\System32> ping 127.0.0.1
```

Pinging 127.0.0.1 with 32 bytes of data:

Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

3.ping 192.168.100.1

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\perehan_sewid>ping 192.168.100.1

Pinging 192.168.100.1 with 32 bytes of data:
Reply from 192.168.100.1: bytes=32 time=2ms TTL=64
Reply from 192.168.100.1: bytes=32 time=2ms TTL=64
Reply from 192.168.100.1: bytes=32 time=3ms TTL=64
Reply from 192.168.100.1: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Users\perehan_sewid>
```

3.packet tracer topology

R1

PhysicalConfigCLIAttributes

```
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up

R1(config-if)#int g0/2
R1(config-if)#ip add 201.191.20.254 255.255.255.0
R1(config-if)#description ##to SW3##
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed
state to up
end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip int br
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0       15.255.255.254  YES manual  up
GigabitEthernet0/1       182.98.255.254  YES manual  up
GigabitEthernet0/2       201.191.20.254  YES manual  up
Vlan1                    unassigned      YES unset   administratively down down
R1#show running config
```

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Top

Cisco Packet Tracer - C:\Users\perehan_sewid\Downloads\Day 08 Lab - IPv4 Addresses.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical 157, y 241

PC1

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 182.98.0.1

Pinging 182.98.0.1 with 32 bytes of data:
Request timed out.
Reply from 182.98.0.1: bytes=32 time=127ms TTL=127
Reply from 182.98.0.1: bytes=32 time=127ms TTL=127
Reply from 182.98.0.1: bytes=32 time=9ms TTL=127

Ping statistics for 182.98.0.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 3ms

C:\>ping 201.191.20.1

Pinging 201.191.20.1 with 32 bytes of data:
Request timed out.
Reply from 201.191.20.1: bytes=32 time=127ms TTL=127
Reply from 201.191.20.1: bytes=32 time=127ms TTL=127
Reply from 201.191.20.1: bytes=32 time=9ms TTL=127

Ping statistics for 201.191.20.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 3ms

C:\>
```

R1

CLI

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Software clause at DFARS sec. 232.227-7013.

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Logical Physical 1828, y 363

PC1

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 182.98.0.1

Pinging 182.98.0.1 with 32 bytes of data:
Request timed out.
Reply from 182.98.0.1: bytes=32 time=127ms TTL=127
Reply from 182.98.0.1: bytes=32 time=127ms TTL=127
Reply from 182.98.0.1: bytes=32 time=9ms TTL=127

Ping statistics for 182.98.0.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 3ms

C:\>ping 201.191.20.1

Pinging 201.191.20.1 with 32 bytes of data:
Request timed out.
Reply from 201.191.20.1: bytes=32 time=127ms TTL=127
Reply from 201.191.20.1: bytes=32 time=127ms TTL=127
Reply from 201.191.20.1: bytes=32 time=9ms TTL=127

Ping statistics for 201.191.20.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 3ms

C:\>
```

R1

CLI

```
Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#do show ip interface brief
Interface IP-Address OK? Method Status
Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
VLAN1 unassigned YES unset administratively down down
R1(config)#interface g0/0
R1(config-if)#ip add 15.255.255.254 255.255.0.0
R1(config-if)#description #to SW1#
R1(config-if)#no shutdown
R1(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up
R1(config-if)#int g0/1
R1(config-if)#ip add 182.98.255.254 255.255.0.0
R1(config-if)#end
R1(config-if)#description #to SW2#
R1(config-if)#no shutdown
R1(config-if)#

```

```
R1#sh run
Building configuration...

Current configuration : 816 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524NMDP-
!
!
```

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Top

```
!
!
interface GigabitEthernet0/0
 description ##to SW1##
 ip address 15.255.255.254 255.0.0.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 description ##to SW2##
 ip address 182.98.255.254 255.255.0.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 description ##to SW3##
 ip address 201.191.20.254 255.255.255.0
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
```

```
R1#  
R1#  
R1#  
R1#wr  
Building configuration...  
[OK]  
R1#
```

 [Top](#)